

# $J/\psi$ radiative corrections — status report

16–17 September 2026

## 1 The final formula is missing from the NLO paper

The NLO paper gives  $\delta_{self}$  (8),  $\delta_{vert}$  (12),  $\delta_{soft}$  (13),  $\delta_{hard}$  only as an unevaluated integral (14), and the exact Bardin–Shumeiko result (28). **There is no compact closed form for the total ultrarelativistic correction** — the quantity an experiment actually applies. The right one is Eq. (24)/(26) of Ehlotzky & Mitter with the missing term restored:

$$\delta(m_V, \omega_{cut}) = -\frac{4\alpha}{\pi} \left[ \left( \frac{1}{2} - \ln \frac{m_V}{m_\ell} \right) \left( \ln r + \frac{1}{3} \right) + \frac{1}{2} \left( L_2(r) - \frac{\pi^2}{6} \right) + \frac{17}{72} + R(r) + \underbrace{\frac{r^2}{16} - \frac{3}{8}r}_{\text{missing from [10]}} \right]$$

$$r = \frac{2\omega_{cut}}{m_V}, \quad R(r) = \left( r - \frac{r^2}{4} - \frac{3}{4} \right) \left[ \ln \left( \frac{m_V}{m_\ell} \sqrt{1-r} \right) - \frac{1}{2} \right], \quad L_2(x) = - \int_0^x \frac{\ln(1-t)}{t} dt$$

For muons add the electron-loop vacuum polarization =  $\delta_{self}$  of Eq. (8) with  $m_e$ . Equivalently the missing piece is  $+(\alpha/\pi)(3r/2 - r^2/4)$  added to  $\delta$ ; it vanishes as  $r \rightarrow 0$  and reaches  $5\alpha/4\pi = 0.0029$  at  $r = 1$ .

## 2 The missing term is Ehlotzky’s error, not yours

Your Eqs. (14), (15), (19), (20) are *faithful transcriptions* of Ehlotzky & Mitter (E&M) Eqs. (24) and (26):

$$\begin{array}{ccccccc} \text{VPK (11)} \leftrightarrow \text{E\&M (22)} & \text{VPK (13)} \leftrightarrow \text{E\&M (23)} & \text{VPK (14)} \leftrightarrow \text{E\&M (24)} & \text{VPK (18)} \leftrightarrow \text{E\&M (25)} \\ & & \text{VPK (19)} \leftrightarrow \text{E\&M (26)} & \end{array}$$

E&M Eq. (23), the differential spectrum, is correct. The term was lost when that spectrum was integrated analytically in 1968. Their Eq. (26) and their Table I inherit it. Recomputing their Table I from their own Eq. (26) reproduces **every printed digit**, electrons and muons:

| $r_{max}$ | E&M $\delta$ | Eq. (26) as published | corrected |
|-----------|--------------|-----------------------|-----------|
| 0.10      | −0.087       | −0.087                | −0.087    |
| 0.25      | −0.034       | −0.034                | −0.033    |
| 0.50      | −0.002       | −0.002                | +0.000    |
| 1.00      | +0.021       | +0.021                | +0.024    |

The error has stood for 57 years. Your paper is the first to correct it.

### 3 Table I of the NLO paper: $m_V = 1.5$ GeV is correct

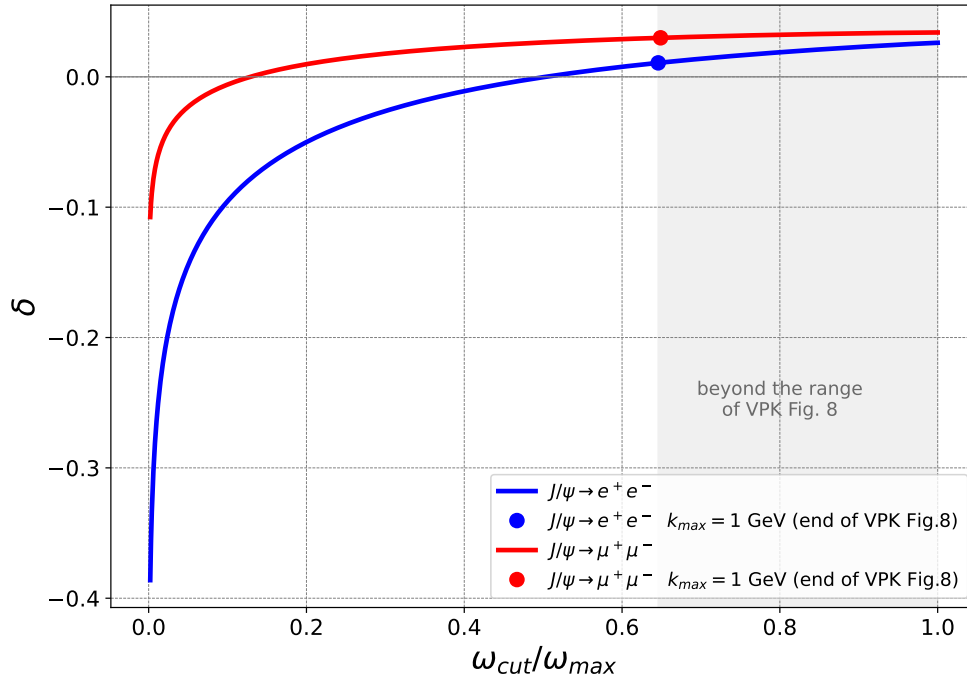
You suspected it was wrong; it is not. E&M's table column is  $2E$  (MeV), and  $2E = m_V$  because  $E = m_V/2$  at resonance (stated below their Eq. 22). The rows are marked  $\rho$  at 765.0,  $\omega$  at 782.0,  $\phi$  at 1019.5 — meson masses. So the last row, 1500.0, is  $m_V = 1.5$  GeV, exactly what the NLO paper used.  $\delta_R$  settles it, depending on the mass only through  $\ln(2E/m)$ :

| reading                             | $\delta_R$ |                               |
|-------------------------------------|------------|-------------------------------|
| $m_V = 1.5$ GeV (NLO's choice)      | -0.746525  | $\leftarrow$ E&M print -0.747 |
| $m_V = 3.0$ GeV (if 1500 were $E$ ) | -0.868832  |                               |
| $m_V = M(J/\psi)$                   | -0.874655  |                               |

What *is* wrong is their claim of agreement with that table: at  $r_{max} = 1$  they have 0.023891 against E&M's 0.021. They agree only in the small- $r_{max}$  rows.

### 4 NLO Fig. 4 vs VPK Fig. 8

Same quantity, different axes.



(a) **The endpoints differ.** NLO uses  $\omega_{max} = m_V/2 - 2m_\ell^2/m_V$  (their Eq. 15); you use  $k_{max} = M_V/2$ . Identical for electrons, but for muons 1.54125 vs 1.54846 GeV, i.e.  $r_{max}^{max} = 0.99535$ , not 1 — a 0.47% shift.

(b) **Your Fig. 8 covers only the first two thirds, stopping at  $k_{max} = 1.0$  GeV =  $0.646\omega_{max}$ .** Their Fig. 4 runs to 1.

| $\omega_{cut}/\omega_{max}$ | $k_{max}(e)$ | $\delta(e)$ | $k_{max}(\mu)$ | $\delta(\mu)$ |
|-----------------------------|--------------|-------------|----------------|---------------|
| 0.10                        | 0.1548       | -0.095444   | 0.1541         | -0.005934     |
| 0.25                        | 0.3871       | -0.036636   | 0.3853         | +0.014282     |
| 0.50                        | 0.7742       | -0.000275   | 0.7706         | +0.026393     |
| 1.00                        | 1.5485       | +0.026135   | 1.5412         | +0.034023     |

At the endpoint against their exact Bardin–Shumeiko numbers:  $e^+e^-$  yours 0.026135 vs exact 0.026135 (diff  $10^{-7}$ );  $\mu^+\mu^-$  yours 0.034017 vs exact 0.034097 (diff  $-8 \times 10^{-5}$ , the expected URA limit).

## 5 New finding: your $k_{max} = M_V/2$ contradicts your own Eq. (22)

The text above your Eq. (16) says the maximum photon energy is  $k_{max} = M_V/2$ . That is the massless limit. The exact endpoint is

$$\omega_{max} = \frac{M_V^2 - 4m_\ell^2}{2M_V} = \frac{M_V}{2} - \frac{2m_\ell^2}{M_V},$$

which is NLO Eq. (15) — **and it is already in your paper**, as Eq. (22),  $E_\gamma = (M_{J/\psi}^2 - M_{ee}^2)/(2M_{J/\psi})$  evaluated at  $M_{ee} \rightarrow 2m_\ell$ . So Eqs. (16) and (22) of your own paper disagree. No consequence for electrons. For muons  $r_{max}$  can only reach 0.99535, so the  $r_{max} = 1$  row of a muon table lies outside phase space. **Not fixed — your call.**

## 6 Formula-by-formula status

| VPK       | NLO            | verdict                                                                                |
|-----------|----------------|----------------------------------------------------------------------------------------|
| (1)–(5)   | (3)–(5)        | agree; conventions differ (fixed $\lambda$ vs polarization-averaged)                   |
| (6)       | (1)+(2)        | agree exactly. <b>Fixed today:</b> $4m_\ell^2/m_V \rightarrow 4m_\ell^2/m_V^2$         |
| (7)       | (6)            | identical (sympy: difference 0)                                                        |
| (8)       | —              | ultrarelativistic limit; NLO has none                                                  |
| (11),(12) | (8)+(12)+(13)  | identical to machine precision                                                         |
| (13)      | (14) integrand | identical                                                                              |
| (14),(15) | —              | were missing the term. <b>Fixed today</b>                                              |
| (16)      | (15)           | <b>disagree</b> — see Section 5                                                        |
| (19),(20) | —              | were missing the term. <b>Fixed today;</b> Eq. (20) $m_e \rightarrow m_\mu$ also fixed |
| (21),(22) | —              | no counterpart; (22) is equivalent to NLO (15)                                         |
| —         | (16)           | FSIC $3\alpha/4\pi$ ; your corrected Eq. (19) reproduces it                            |
| —         | (28)           | exact Bardin–Shumeiko; no counterpart in your paper                                    |

## 7 Changed today

Paper (Jpsi\_RC.tex, compiled and published): Eqs. (14),(15),(19),(20) missing term added; Eq. (20) leading log  $m_e \rightarrow m_\mu$ ; Eq. (6)  $4m_\ell^2/m_V \rightarrow 4m_\ell^2/m_V^2$ ; Fig. 7 recomputed at  $\varepsilon = 0.001$  (was 0.01, caption said 0.001); Figs. 3–8 in-plot titles removed, information moved to captions; Figs. 3,4,6 captions now name the two decay modes.

Code: three duplicate copies collapsed into one library `rc_vector_mesons.py`; `validate_rc.py` checks it against an independent quadrature reference and two analytic anchors — all pass. Repo `~/jpsi-rc-2026`, <https://github.com/vkubarovsky/jpsi-rc-2026>, mirrored to OneDrive Work/2026\_Jpsi\_rad

## 8 Waiting on you

- `notes/NLO_comments.pdf` — the note for your co-authors: all six of your items plus the Eq. (4) polarization-sum point. *Not sent anywhere.*
- Section 5: Eq. (16) vs Eq. (22) inconsistency — not fixed.
- “Formala”  $\times 2$  in the Fig. 9 legend — you said leave Fig. 9 alone.
- `fig_igor` defined twice in `Jpsi_rad_corr.py`.
- Never verified: your Eqs. (2)–(5) in detail, (17), (18); and the Bethe–Heitler formulas behind the  $-20\%$  claim in Section VII.